METRIC

DOD-P-15146B
21 November 1977
SUPERSEDING
MIL-P-15146A
27 June 1951
(See 6.7)

MILITARY SPECIFICATION

PAINT, EXTERIOR, DULL BLACK (FORMULA NO. 104) (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dull black exterior paint (Formula No. 104) for use on surface ships.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
TT-R-266 - Resin, Alkyd Solutions.
TT-T-291 - Thinner-Paint, Volatile Spirits, Petroleum Spirits.
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

MILITARY
MIL-P-15173 - Pigment, Magnesium Silicate, Dry (Paint Pigment).
MIL-S-15191 - Silica, Diatomaceous (Flatting-extender Pigment).

STANDARDS

FEDERAL
FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Method of Inspection, Sampling, and Testing.
FED-STD-313 - Material Safety Data Sheets, Preparation and the Submission of.
FED-STD-595 - Colors.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8010

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 93 - Flash Point by Pensky-Martens Closed Tester, Test for.
D 209 - Lampblack, Spec. for.
D 562 - Consistency of Paints Using the Stormer Viscosimeter, Test for.
D 563 - Phthalic Anhydride Content of Alkyd Resins and Resin Solutions, Test for.
D 600 - Liquid Paint Driers, Spec. for.
D 1210 - Fineness of Dispersion of Pigment-Vehicle Systems, Test for.
D 1296 - Odor of Volatile Solvents and Diluents, Test for.
D 1306 - Phthalic Anhydride Content of Alkyd Resins and Esters Containing Other Dibasic Acids (Gravimetric), Test for.
D 1308 - Effect of Household Chemicals on Clear and Pigmented Organic Finishes, Test for.
D 1475 - Density of Paint, Varnish, Lacquer, and Related Products, Test for.
D 1542 - Rosin in Varnishes, Qualitative Tests for.
D 2369 - Volatile Content of Paints, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.6). Questions pertinent to this effect shall be referred by the procuring activity to the appropriate service medical department which will act as advisor to the procuring activity.

3.2 Formula. The paint shall consist of ingredients proportioned as specified (see table I), except that small quantities of antisettling, antisagging, and antiskinning agents may be added to the formulation, provided that all other requirements of the specification are met. Test reports required by method 1031 of FED-STD-141 shall include the exact formula used.

TABLE I. Formula No. 104.

Ingredients	Specification	kg[1/]	(Pounds)[1/]
Lampblack	ASTM D 209	29.5	(65)
Diatomaceous silica	MIL-S-15191	22.7	(50)
Magnesium silicate	MIL-P-15173, type A	88.4	(195)
Alkyd resin solution	TT-R-266, type I, class A[2/3/]	163.5	(360)
Paint thinner	TT-T-291, type II, grade A[3/]	122.0	(270)
Lead naphthenic drier	ASTM D 600, class B	4.52	(10)
Cobalt naphthenic drier	ASTM D 600, class B	0.543	(1.2)
Manganese naphthenic drier	ASTM D 600, class B	0.543	(1.2)

[1/] The formula is given slightly in excess of 100 gallons to allow for normal manufacturing loss.

[2/] The solvent shall consist of mineral spirits conforming to type II, grade A of TT-T-291.

[3/] Alkyd resin solution conforming to type I, class B of TT-R-266 may be substituted, provided that the weight of the alkyd resin solution is increased from 163.636 kilograms (kg) to 191.364 kg (360 to 421 pounds) and weight of paint thinner reduced from 122.727 kg to 101.818 kg (270 to 224 pounds).

3.2.1 The formula shown in table I is designated Navy Standard Formula No. 104. Whenever Formula No. 104 is specified, the paint shall conform to this specification (see 6.6).

3.3 Manufacture. The component raw materials shall be mixed and ground as required to produce a product which is uniform, free from dirt and grit, and in full conformity to the requirements of this specification.

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3.4 Quantitative requirements. The paint shall conform to the quantitative requirements in table II when tested as specified in accordance with 4.4.

TABLE II. Quantitative requirements.

Characteristics	Requirements	
	Minimum	Maximum
Pigment, percent by weight of paint	31.5	35.5
Volatiles, percent by weight of paint	38.5	42.5
Nonvolatile vehicle, percent by weight of paint (calculated by difference)	26.0	30.0
Phthalic anhydride, percent by weight of nonvolatile vehicle	23.0	----
Water, percent by weight of paint	----	0.5
Coarse particles and skins (as residue retained on No. 325 sieve), percent by weight of paint	----	0.5
Viscosity, Krebs units	70	80
Weight per gallon, kg (pounds)	4.1 (9.0)	4.4 (9.6)
Drying time - set-to-touch, hours	----	1
dry hard, hours	----	5
Gloss, 60-degree specular	----	5
Fineness of grind	4	----
Flash point, °C (°F)	38 (100)	----
Lampblack, percent by weight of pigment	18	----
Calcium (as CaO), percent by weight of pigment	----	6
Aluminum and iron (as R_2O_3), percent by weight of pigment	----	2.5
Barium (as $BaSO_4$), percent by weight of pigment	----	None

3.4.1 Solvent. The solvent portion of the formulation shall conform to requirements herein specified.

(a) A combination of hydrocarbons, alcohols, aldehydes, ethers, esters, or ketones having an olefinic or cycloolefinic type of unsaturation except perchloroethylene: 5 percent maximum.
(b) A combination of aromatic compounds with eight or more carbon atoms to the molecule except ethylbenzene, methyl benzoate, and phenyl acetate: 8 percent maximum.
(c) A combination of ethylbenzene, ketones having branched hydrocarbon structures, trichloroethylene, or toluene: 20 percent maximum.

3.5 Qualitative requirements. The paint shall conform to the qualitative requirements specified herein.

3.5.1 Odor. The odor shall be normal for the volatiles permitted when tested as specified (see 4.4).

3.5.2 Color. The color of the paint shall match the dry color chip No. 37038 of FED-STD-595 when tested as specified (see 4.4.6).

3.5.3 Flexibility. The paint shall show no evidence of cracking when tested as specified (see 4.4.14).

3.5.4 Compatibility with thinner. There shall be no evidence of incompatibility of any of the ingredients of the paint as received when tested as specified (see 4.4.5).

3.5.5 Condition in container. The paint, as received in its filled, original container, shall be readily mixable, either by paddle, shaker, or mechanical stirrer, to a smooth, uniform consistency and shall conform to all requirements related to the finished product as listed in table II. The paint shall not increase more than 10 Krebs units in viscosity or increase more than 2 hours in drying time. It shall not curdle, liver, gel, seed, or develop any other objectionable properties for a minimum of 1 year after date of manufacture, when tested as specified (see 4.4.4).

3.5.6 Skinning. The paint shall show no skinning after 48 hours in a tightly closed, partially filled container when tested as specified (see 4.4.7).

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3.5.7 Resistance to water immersion. There shall be no evidence of whitening, blistering, softening, or loss of adhesion when tested by the method specified (see 4.4.13).

3.5.8 Rosin and rosin derivatives. Rosin and rosin derivatives shall not be present when tested as specified (see 4.4.10).

3.5.9 Phenolic resins. Phenolic resins shall not be present when tested as specified (see 4.4.11).

3.5.10 Application characteristics.

3.5.10.1 Brushing properties. The paint, when tested as specified (see 4.4.8), shall be capable of being brushed out and laid off without excess drag on the brush. When dry, the brush-coated surface shall be free from sags, runs, wrinkles, excess brush marks, or other film defects. The film shall exhibit good adhesion and a smooth, uniform appearance.

3.5.10.2 Spraying properties. The paint, when tested as specified (see 4.4.9), shall show no running, sagging, streaking, dusting, mottling, color separation, or any other film defects. The film shall exhibit good adhesion and a smooth, uniform appearance.

3.5.11 Material Safety Data Sheet. The procuring activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is DD Form 1813 and found in and part of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall be provided in accordance with method 1031 of FED-STD-141.

4.3 Additional inspection.

4.3.1 Ingredient materials. When requested by the testing laboratory or other controlling authority, 1 pint of each ingredient in the formula specified in table I shall be supplied for test purposes (see 6.2).

4.4 Test procedures. Tests shall be conducted in accordance with methods specified (see table III).

TABLE III. Test procedures.

Tests	Applicable method in FED-STD-141	Applicable ASTM test method
Pigment content	4021	----
Volatiles	----	D 2369
Nonvolatile vehicle (by difference)	4053	----
Phthalic anhydride	----	D 563
Drying time	4061	----
Water	4081	----
Coarse particles and skins	4092	----
Weight per gallon	----	D 1475
Compatibility with thinner	4203	----
Color	4250	----
Viscosity, Krebs-Stormer	----	D 562
Flash point	----	D 93
Brushing properties	4321	----

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TABLE III. Test procedures. - Continued

Tests	Applicable method in FED-STD-141	Applicable ASTM test method
Spraying properties	4331	----
Odor	----	D 1296
Fineness of grind	----	D 1210
Rosin and rosin derivatives	----	D 1542, Sect. 4
Phenolic resins	5141	----
Water immersion	----	D 1308 Sect. 5
Gloss	6101	----
Flexibility	6221	----
Condition in container	3011	----
Skinning	3021	- --

4.4.1 Pigment analysis.

4.4.1.1 Pigment content. Extract the pigment from a weighed sample of paint as in method 4021 of FED-STD-141, using extraction mixture A. Dry and weigh extracted pigment. Calculate percent pigment in the paint.

4.4.1.2 Lampblack. Transfer 1 gram(g) of the extracted pigment to a porcelain crucible. Ignite for 10 minutes in a current of dry carbon dioxide (using a Rose crucible cover) with a medium flame from a Meker burner. Cool in a current of dry carbon dioxide and weigh. Then ignite with free access of air until all carbon is consumed. Cool and weigh. The difference in weight between the second and final weighings is calculated as lampblack.

4.4.1.3 Aluminum and iron. Weigh 1 g of the extracted pigment into a platinum crucible. Ignite at 730°C (1350°F) to remove all carbon pigment or organic matter. Allow to cool, moisten with water, and add 5 drops of sulfuric acid (1:1) and 15 milliliters (mL) of hydrofluoric acid. Evaporate on a steam bath until only sulfuric acid remains. Transfer the crucible to a sand bath and heat cautiously to expel the sulfuric acid. Cool and fuse the residue with 1 g of sodium carbonate, using a well-fitting lid on the crucible. Start the fusion over a small flame and increase the heat gradually until the entire contents of the crucible are in a molten state. Continue heating 20 minutes. When the fusion is completed, allow the crucible and contents to cool and transfer to a 600-mL beaker containing about 200 mL of water. Boil until the melt is disintegrated. Remove the crucible and lid, being careful to scrub out any adhering particles of melt. Carefully acidify the contents of the beaker with concentrated hydrochloric acid, introducing the acid in small portions under a watch crystal which covers the beaker in order to avoid loss by spattering. Add 3 or 4 mL of acid in excess and boil to expel carbon dioxide. Add 2 or 3 drops of sulfuric acid (1:1). At this point, the solution should be clear. (If a precipitate is present, a qualitative test for barium shall be made.) Add methyl orange indicator and neutralize with ammonium hydroxide and then add 3 or 4 drops in excess. Boil 3 or 4 minutes and set aside for 5 minutes. Then filter through a fast filter paper and wash with water made just alkaline to methyl orange with ammonium hydroxide. Ignite the precipitate to constant weight at 870°C (1600°F). Calculate as percent mixed aluminum and iron oxides.

4.4.1.4 Calcium. Adjust the volume of filtrate from the aluminum and iron determination to about 300 mL and make slightly acidic with hydrochloric acid. Slowly add 25 mL of saturated solution of ammonium oxalate, stirring continuously. Add 40 mL of concentrated ammonium hydroxide; boil 5 minutes and set on the edge of the steam bath overnight. Cool to room temperature. Filter off the precipitated calcium oxalate and wash with hot water until all ammonium oxalate is removed (usually 10 to 12 washings). Wash the calcium oxalate back into the beaker in which it was originally precipitated. Add 15 mL of sulfuric acid (1:1); dilute to 300 mL. Spread the open filter paper on the side of the beaker above the solution. Heat the solution to 80°C (176°F) and titrate with N/10 $KMnO_4$ until a pink color appears. Now quickly immerse the filter paper and complete the titration.

1 mL of N/10 $KMnO_4$ = 0.0028g CaO

4.4.1.5 Barium. Conduct test for barium in accordance with standard methods of qualitative analysis.

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4.4.2 Phthalic anhydride. Determine phthalic anhydride content of the nonvolatile vehicle in accordance with ASTM D 563, applying the correction procedure described in 4.2 of the test method. If dibasic acids other than phthalic are present, ASTM D 1306 shall be used.

4.4.3 Drying time. Determine drying time by method 4061 of FED-STD-141, except that the specified conditions of temperature and humidity shall apply only for referee tests in case of dispute. All other tests shall be conducted under prevailing laboratory conditions.

4.4.4 Condition in container. The contractor shall determine package condition of paint in accordance with method 3011 of FED-STD-141 and observe for compliance with 3.5.5. The Government, at its option and at any time not to exceed 1 year after manufacture, may test paint stored in its original containers for product condition, viscosity, and dry hard time. (Any action by the Government to disqualify paint after prior acceptance shall be based on the examination of paint stored in its original containers.)

4.4.5 Compatibility with thinner. Determine compatibility with thinner in accordance with method 4203 of FED-STD-141. Fifty mL of paint shall be mixed with 50 mL of mineral spirits conforming to type II, grade A of TT-T-291. Observations shall be made immediately after mixing and repeated in 30 minutes.

4.4.6 Color. Prepare test panel by applying a single drawdown coat of the paint to a planar piece of opaque white glass using a doctor blade with a clearance of 0.0150 centimeters (cm) (0.006 inch) [designed to give a wet film thickness of approximately 0.0075 cm (0.003 inch)]. The coated panel shall be allowed to dry 24 hours under prevailing laboratory conditions before comparing with the standard color card as specified in method 4250 of FED-STD-141 (see 3.5.2).

4.4.7 Skinning. Determine skinning characteristics of the material in a partially filled container in accordance with method 3021 of FED-STD-141.

4.4.8 Brushing properties. Determine brushing properties of the paint in accordance with method 4321 of FED-STD-141 and observe for compliance with 3.5.10.1.

4.4.9 Spraying properties. Determine spraying properties of the paint in accordance with method 4331 of FED-STD-141. Reduce eight parts by volume of paint with one part by volume of thinner conforming to TT-T-291, type II, grade A. Spray coat the steel panel to a dry film thickness of 0.0025 cm ± 0.00025 cm (0.001 ± 0.0001 inch) and observe for compliance with 3.5.10.2. For referee tests, the automatic film application described in method 2131 of FED-STD-141 shall be used.

4.4.10 Rosin and rosin derivatives. Conduct test for rosin and rosin derivatives in accordance with Lieberman-Storch Test of ASTM D 1542, section 4. A portion of the separated, nonvolatile vehicle shall be used for the test.

4.4.11 Phenolic resins. Conduct test for phenolic resins in accordance with method 5141 of FED-STD-141. A portion of the nonvolatile vehicle shall be used for the test.

4.4.12 Gloss. Determine 60-degree specular gloss in accordance with section 5 of ASTM D 1308. Prepare the panels for test by applying a single drawdown coat of paint using a doctor blade designed to give a dry film thickness of 0.0025 cm ± 0.00075 cm (1.0 ± 0.3 mils). Dry the panels for 48 hours under prevailing laboratory conditions of temperature and humidity in a room free from dust and fumes.

4.4.13 Water immersion test. Determine water immersion properties of the paint by immersion method of section 5 of ASTM D 1308. Apply the paint coating to a wet film thickness of 0.0037 cm ± 0.00050 cm (0.0015 ± 0.0002 inch). Allow 24 hours for the coating to air dry; then immerse it in distilled water at 25°C ± 5°C (77°F ± 9°F) for 18 hours. After removal from the water, allow a 2-hour recovery period before examination for compliance with 3.5.7.

4.4.14 Flexibility. Determine flexibility in accordance with method 6221 of FED-STD-141. Draw down a 5.08-cm (2-inch) wide film of the paint with a suitable film applicator that will give a dry film thickness of 0.0025 cm ± 0.0076 cm (0.0010 ± 0.0003 inch) on a flat tin panel prepared in accordance with method 2012 of FED-STD-141, using the aliphatic naphtha-ethylene glycol monoethyl ether mixture. Air dry the test specimen for 2 hours in a horizontal position; then bake for 24 hours in an air-circulating oven at 100°C to 105°C (212°F to 220°F). At the end of the baking period, condition the panel for 30 minutes at 25°C ± 5°C

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(77°F + 9°F) and bend over a 0.32-cm (1/8-inch) mandrel. The coated surface of the panel shall be uppermost during the bending, which shall be accomplished at a uniform rate over approximately 1 second. The panel shall be examined at the bend using a seven-power lens and any cracking shall be noted.

4.5 Inspection of preparation for delivery. Inspect the packaging, packing, and marking of the material to determine compliance with the requirements of Section 5 of this specification.

4.6 Toxicity. A manufacturer of material shall disclose the formulation of his product to the Navy Bureau of Medicine and Surgery, Navy Department, Washington, DC 20372. The disclosure of proprietary information, which shall be held in confidence by the Bureau of Medicine and Surgery, shall include: the name, formula, and approximate percentage by weight and volume of each ingredient in the product; the results of any toxicological testing of the product; identification of its pyrolysis products; and any such other information as may be needed to permit an accurate appraisal of any toxicity problem associated with the handling, storage, application, use, or disposal of the material.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Packaging, packing, and marking. The paint shall be packaged, packed, and marked in accordance with PPP-P-1892. The level of packaging shall be A or C and the level of packing shall be A, B, or C as specified (see 6.2). The paint shall be furnished in 1-gallon cans or 5-gallon pails as specified (see 6.2).

5.1.1 Special marking. In addition to the markings required by the contract or order (see 6.2), each container, interior and exterior, shall be marked with the following:

"The volatile content of the material in this container is not photochemically reactive as defined by Rule 102 of the South Coast Air Quality Management District." (See 6.5).

6. NOTES

6.1 Intended use. This paint is intended for use on the exterior of surface ships and may be applied where air pollution regulations apply.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Special samples if required (see 4.3.1).
(c) Level of packaging and level of packing required (see 5.1).
(d) Size of container (see 5.1).
(e) Special marking required (see 5.1.1).

6.3 Packaging. Level B is intended to provide economical but limited protection and should be specified only when it is determined the paint will be held in covered storage no more than 1 year from date of initial packaging.

6.4 Basis of purchase. The enamel should be purchased under this specification by volume, the unit being the U.S. gallon at 15.5°C (60°F).

6.5 Volatile content. Although the container marking specifically refers to the South Coast Air Quality Management District, the paint may be used anywhere else paint complying with 3.4.1 is allowed. This includes all other air pollution control districts or similar areas controlling this emission of solvents into the atmosphere. Information regarding Los Angeles County Air Pollution Rules 102, 442, and 443 may be obtained from: South Coast Air Quality Management District, Metropolitan Zone, 434 South San Pedro Street, Los Angeles, California 90013.

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6.6 Composition by volume. For information only and with the understanding that weight-volume relationships of ingredients may vary slightly, the following approximation of composition by volume is included.

TABLE IV. Composition by volume.

Ingredients	Gallons
Lampblack	4.38
Diatomaceous silica	3.00
Magnesium silicate	8.21
Alkyd resin solution	45.0 (28.64) 1/
Paint thinner	40.60
Lead naphthenic drier	1.02
Cobalt naphthenic drier	0.16
Manganese naphthenic drier	0.16
Total volume	102.53

1/ Figure in parentheses refers to volume of resin solids (nonvolatile).

6.7 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
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Preparing activity:
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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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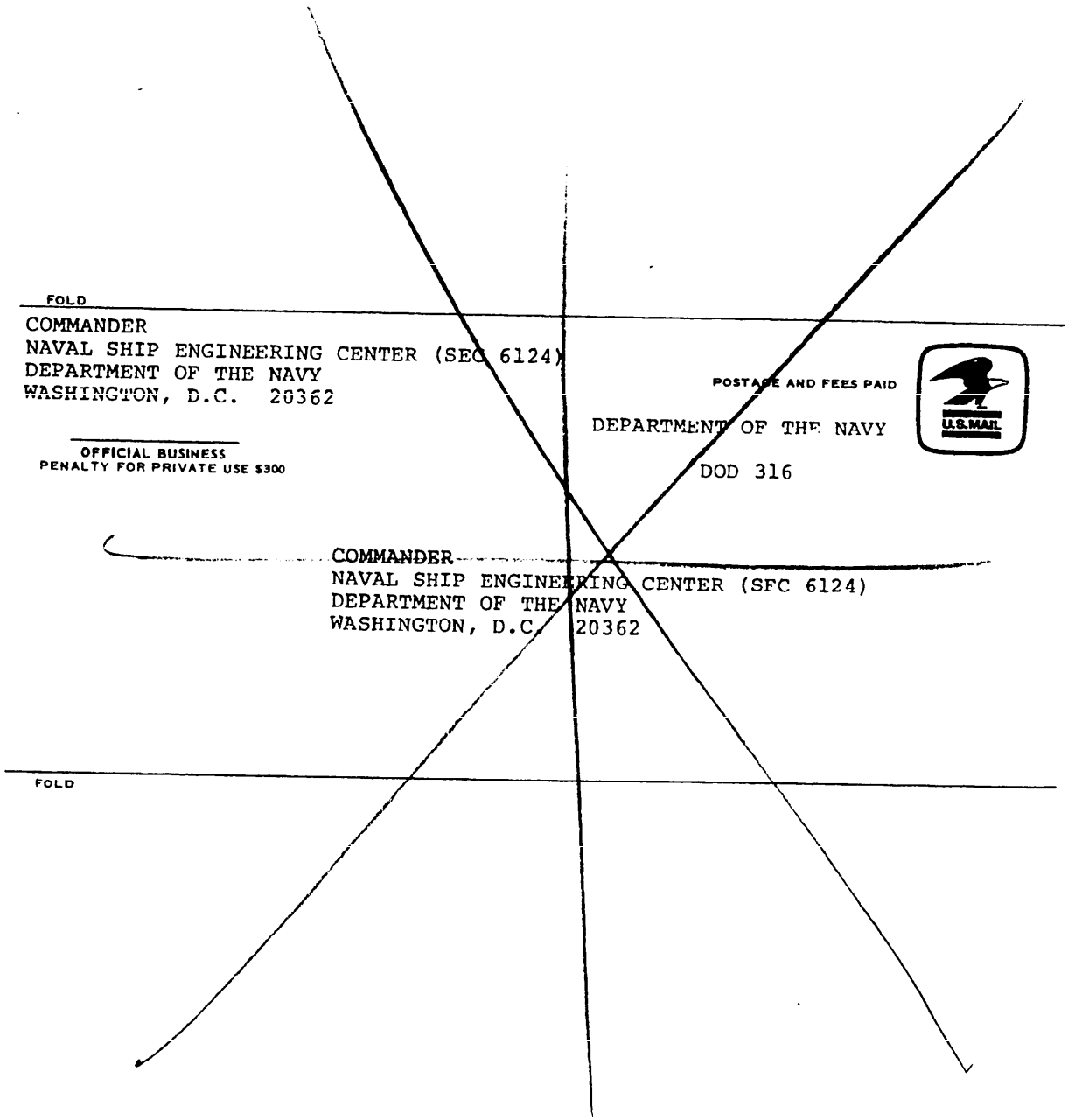
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